Chapter 12, Problem 52P

Problem

A four-wire wye-wye circuit has

$$\mathbf{V}_{an} = 220\underline{/120^\circ}, \qquad \mathbf{V}_{bn} = 220\underline{/0^\circ}$$

$$\mathbf{V}_{cn} = 220\underline{/-120^\circ}\ \text{V}$$

If the impedances are

$$\mathbf{Z}_{AN} = 20\underline{/60^\circ}, \qquad \mathbf{Z}_{BN} = 30\underline{/0^\circ}$$

$$\mathbf{Z}_{cn} = 40\underline{/30^\circ}\ \Omega$$

find the current in the neutral line.

Step-by-step solution

Step 1 of 5

Consider the circuit diagram shown in Figure 1.

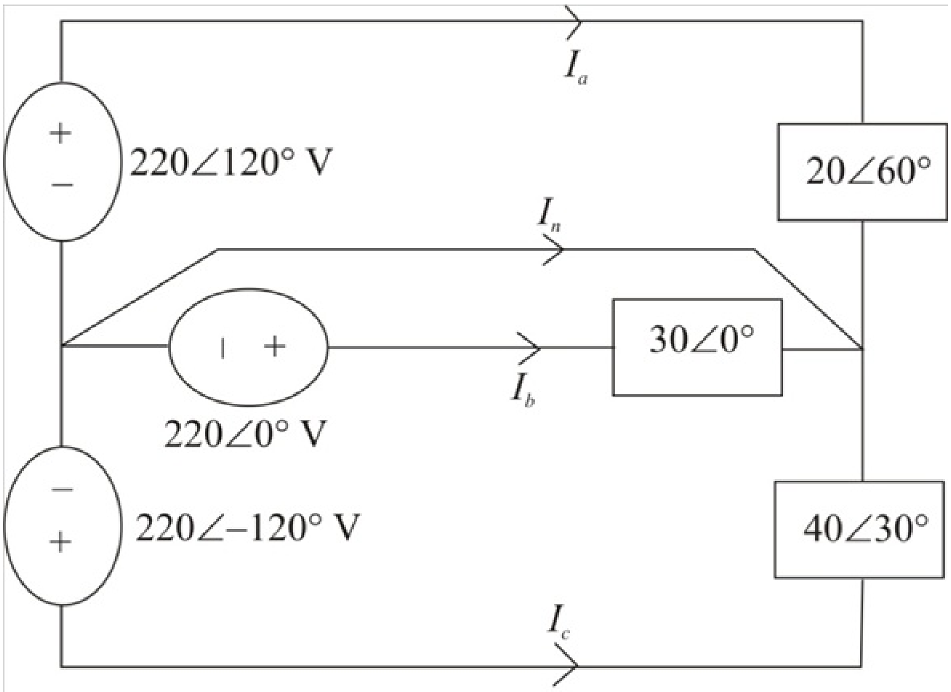

Figure 1

Step 2 of 5

Calculate the value of the current I_a.

$$I_a = \frac{220\angle 120^\circ}{20\angle 60^\circ}$$
$$= 11\angle 60^\circ\ \text{A}$$

Therefore, the value of the current I_a is, $11\angle 60^\circ\ \text{A}$.

Step 3 of 5

Calculate the value of the current I_b.

$$I_b = \frac{220\angle 0^\circ}{30\angle 0^\circ}$$
$$= 7.33\angle 0^\circ \text{ A}$$

Therefore, the value of the current I_b is, $7.33\angle 0^\circ \text{ A}$.

Step 4 of 5

Calculate the value of the current I_c.

$$I_c = \frac{220\angle -120^\circ}{40\angle 30^\circ}$$
$$= 5.5\angle -150^\circ \text{ A}$$

Therefore, the value of the current I_c is, $5.5\angle -150^\circ \text{ A}$.

Step 5 of 5

Calculate the value of the current I_n.

$$I_n = -(I_a + I_b + I_c)$$

Substitute $11\angle 60^\circ \text{ A}$ for I_a, $7.33\angle 0^\circ \text{ A}$ for I_b and $5.5\angle -150^\circ \text{ A}$ for I_c.

$$I_n = -(11\angle 60^\circ + 7.33\angle 0^\circ + 5.5\angle -150^\circ)$$
$$= -(10.54\angle 40^\circ)$$
$$= 10.54\angle -140^\circ \text{ A}$$

Therefore, the value of the current I_n is, $\boxed{10.54\angle -140^\circ \text{ A}}$.